PES UNIVERSITY

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# “MODELING RELATIONSHIP BETWEEN EMPLOYEE TRAINING HOURS AND PRODUCTIVITY USING REGRESSION”

MBA PROGRAM

RESEARCH-BASED PROJECT

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**Modeling Relationship Between Employee Training Hours and Productivity Using Regression**


# ABSTRACT

Employee training is widely recognized as a crucial driver of organizational productivity, yet empirical evidence on how training hours quantitatively influence employee output remains varied across industries. This study examines the relationship between total training hours completed in a 12-month period and self-reported productivity outcomes using data collected through a structured survey of working professionals. A simple linear regression model was employed under the Classical Linear Regression Model (CLRM) framework, with training hours as the independent variable and post-training productivity rating as the dependent variable. Preliminary descriptive analysis indicated notable variation in both training exposure and productivity levels across respondents. Regression results revealed a positive association between training hours and productivity ratings, supporting the hypothesis that increased training is associated with improved employee performance. The findings contribute empirical support for strategic training investments and highlight the need for organizations to adopt measurable learning interventions. Future research may expand the model by incorporating additional predictors such as job experience, motivation, and work environment to improve explanatory power.



***Keywords:*** *Training hours, Employee productivity, CLRM, Linear regression, Human capital development*

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## INTRODUCTION

In contemporary organizational environments, human capital development has emerged as a critical determinant of sustained competitiveness, operational excellence, and strategic adaptability. As industries evolve due to technological advancement, globalization, and shifting market dynamics, the need for employees to continuously enhance their skills and competencies has become increasingly pronounced. Among the various mechanisms of human capital development, structured training programs represent one of the most widely adopted interventions to improve workforce capabilities, enhance productivity, and align employee performance with organizational goals. Training not only equips employees with updated technical knowledge and job-related proficiencies but also strengthens behavioral, cognitive, and attitudinal dimensions that contribute to improved performance outcomes.

Despite the universal consensus on the importance of training, academic and industry literature reveals substantial variation in how training effectiveness is conceptualized, measured, and evaluated across sectors. Productivity, as a core indicator of performance, is influenced by multiple organizational, individual, and contextual variables, making direct attribution to training challenging. Nonetheless, training is frequently cited as a prerequisite for higher individual output, stronger engagement, reduced task errors, and improved decision-making capabilities. The fundamental premise behind training investments is that enhanced skill sets enable employees to execute tasks more efficiently and effectively, thereby contributing to greater organizational productivity. This aligns with human capital theory, which views the acquisition of skills and knowledge as an economic investment that yields measurable returns in the form of improved performance and productivity.

While numerous conceptual and qualitative studies have examined the broader relationship between training and performance, there is comparatively less empirical work focusing on quantifying how training exposure—measured through training hours—directly influences productivity ratings within a workforce. Many existing studies investigate training qualitatively, explore organizational case studies, or examine cross-sectional industry trends without applying

statistical models that assess causality or predictive behavior. As a result, there exists a research opportunity to quantitatively analyze the relationship between training hours and productivity within a well-structured analytical framework. Such analysis not only contributes to academic literature but also assists organizations in making evidence-based training investments rather than intuitive or compliance-driven decisions.

From a methodological standpoint, modeling productivity outcomes using the Classical Linear Regression Model (CLRM) offers a structured and statistically valid approach for examining training–productivity relationships. The CLRM provides key assumptions regarding linearity, independence, homoscedasticity, and normality that, when satisfied, allow for unbiased estimation and inference regarding the effect of training hours on productivity. By treating training hours as an independent variable and post-training productivity ratings as a dependent variable, regression modeling can reveal the extent to which incremental increases in training hours correspond to measurable improvements in productivity. Such an approach not only quantifies the strength and direction of the relationship but also enables hypothesis testing to determine whether training significantly influences productivity.

The present study aims to contribute to this empirical gap by utilizing primary data collected through a structured survey administered to working professionals across varying age groups, job roles, employment types, and experience levels. The dataset includes variables related to demographic factors, organizational characteristics, training exposure, and productivity outcomes, enabling a detailed exploration of the training–productivity dynamic. Notably, the dataset includes both continuous measures (e.g., total training hours, average tasks completed per week, percentage of targets achieved) and subjective evaluations (e.g., perceived improvement in performance, post-training productivity rating), allowing for a richer interpretation of productivity as both an objective and perceived construct.

Understanding the relationship between training hours and productivity carries significant practical implications. For policymakers and organizational leaders, empirical evidence regarding the value of training can guide budget allocation, training program design, and strategic workforce planning. For employees, structured training may influence career progression, job satisfaction, and perceived self-efficacy. For academic researchers, the

integration of statistical modeling with workplace performance data expands methodological perspectives within the field of human resource development and organizational behavior.

In summary, the increasing emphasis on skill development within dynamic work environments underscores the need to assess the measurable outcomes of training interventions. By applying a regression-based analytical framework under CLRM assumptions, this study seeks to model the relationship between employee training hours and productivity and thereby provide empirical contributions to the broader discourse on human capital development. The insights drawn from this investigation have the potential to enhance academic understanding, support organizational decision-making, and inform future research on employee development and performance optimization.

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## OBJECTIVES

1. To analyse the extent of variation in employee training hours and corresponding productivity levels using descriptive statistics based on primary survey data.

2. To determine whether there exists a statistically significant relationship between total training hours completed and post-training productivity ratings at the employee level.

3. To evaluate the predictive influence of training hours on productivity outcomes utilizing a Classical Linear Regression Model (CLRM) framework.

4. To identify whether incremental increases in training exposure contribute to measurable improvements in individual productivity indicators within diverse workplace settings.

5. To provide empirical insights that can support evidence-based decisions regarding organizational training investments and human capital development strategies.

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## LITERATURE REVIEW

The relationship between employee training and productivity has been widely studied across human resource development, organizational behavior, and managerial economics. Training, conceptually, is viewed as a structured activity aimed at enhancing job-related knowledge, skills, and competencies, thereby enabling employees to perform tasks more efficiently and effectively. Productivity, on the other hand, is commonly associated with output quality, task completion, target achievement, and performance improvement at both individual and organizational levels. The collective body of literature suggests that training represents a strategic human capital investment that produces measurable performance and productivity gains, though the strength and nature of this relationship varies across industries, organizational contexts, and methodological approaches.

Early studies in this domain emphasized the theoretical linkage between human capital development and organizational performance. Drawing from human capital theory, training is positioned as an economic investment where employees acquire knowledge and skills that increase their marginal productivity, ultimately benefitting both the individual and the firm. Subsequent empirical research has attempted to validate this connection through organizational case studies, survey-based analyses, and quantitative modeling techniques. Many of these studies point toward a positive association between training interventions and productivity outcomes. For instance, research examining comparative training practices across industries highlighted that although training methods and intensity differ by sector, well-designed training programs consistently contribute to improved employee performance and task efficiency. Similar findings were reported in cross-sectional studies where organizations investing in structured training observed reductions in workflow errors, improved employee motivation, and increased output quality.

A parallel stream of literature focuses on measuring the impact of training through quantitative and econometric lenses. Studies conducted in manufacturing and service organizations indicate that variables such as training hours, training content, training relevance, and delivery mode

significantly influence productivity metrics such as output-unit production, task completion rate, and target achievement percentage. Regression-based studies, particularly those employing linear and structural equation modeling (SEM), consistently show training as a statistically significant predictor of employee productivity. For example, partial least squares structural modeling has been used to demonstrate that both general training and strategic skills training exert positive effects on employee productivity, suggesting that productivity gains are not only technical but also influenced by cognitive and strategic learning processes.

In addition to technical training, recent literature expands the discussion to broader developmental practices such as career development and organizational learning. Studies focusing on career training demonstrate that employees exposed to continuous learning opportunities report enhanced technical competence, higher job proficiency, and improved performance outcomes. This aligns with research conducted in sectors facing high technological change, where training is considered a mechanism for maintaining workforce relevance and competitive advantage. Conceptual papers also reinforce this perspective by suggesting that firms operating in dynamic competitive environments cannot sustain performance without ongoing training and development investments that enhance employee motivation, innovation capacity, and adaptability.

While most literature converges on a positive association between training and productivity, several studies acknowledge contextual and methodological complexities. A number of authors highlight that the impact of training varies based on organizational size, managerial commitment, resource availability, and the alignment between training content and job requirements. In some cases, research findings show variation in effect size across industries and work roles, implying that training may not uniformly translate into productivity gains without appropriate implementation, evaluation, and follow-up. Moreover, studies conducted in public-sector or resource-constrained environments note challenges such as limited training budgets, inadequate training needs assessment, and lack of standardized evaluation metrics, which collectively weaken the observable impact of training on performance outcomes.

Another important theme in the literature relates to measurement challenges. Productivity is a multidimensional construct that includes objective indicators (e.g., output quantity, task efficiency) as well as subjective indicators (e.g., self-reported productivity, perceived

improvement). Many studies rely on perception-based surveys, which may introduce bias or measurement limitations, while others adopt financial or operational output measures that capture performance more accurately but are harder to collect in cross-industry research. Recent literature attempts to bridge this gap by combining subjective and objective measures, for example through surveys that capture self-reported productivity alongside metrics such as percentage of targets achieved, or units produced per hour. Such mixed-methods approaches provide richer insights into the training–productivity relationship.

Overall, the existing body of knowledge substantiates the theoretical claim that training contributes positively to employee productivity and organizational performance. However, it also highlights the need for empirical modeling approaches that quantify the strength of this impact under specific assumptions. This creates an important opening for studies employing Classical Linear Regression Modeling (CLRM) to statistically assess how variations in training hours influence productivity outcomes. By operationalizing training exposure as a measurable input and productivity as a measurable output, such research addresses both academic and managerial interests, offering evidence-based insights for training investment decisions, workforce planning, and human capital strategy.

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## RESEARCH GAP:

Although extensive literature acknowledges that training enhances employee performance and organizational productivity, a clear empirical gap exists in quantifying how training hours specifically influence productivity outcomes under statistical modeling assumptions. Much of the prior research relies on qualitative assessments, conceptual reviews, industry case studies, and perception-based surveys, which highlight positive associations but do not establish directional causality or measurable effect size. Even studies employing quantitative methods often analyze training as a categorical HR practice rather than a measurable continuous variable, or they evaluate productivity using aggregate organizational indicators rather than employee-level performance data. Additionally, the majority of empirical studies are conducted within single-industry contexts or focus on broad constructs such as job satisfaction, motivation, or performance, leaving limited evidence on how variations in training exposure translate into incremental productivity gains at the individual level. There is also a lack of research applying Classical Linear Regression Model (CLRM) assumptions to assess whether training–productivity relationships hold under statistical conditions such as linearity, independence, and homoscedasticity. Therefore, a methodological and contextual gap persists in understanding whether and to what extent increased training hours predict improved productivity within diverse work settings, thereby necessitating studies that employ regression-based empirical modeling using primary survey data to validate this relationship.

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## RESEARCH METHODOLOGY:

1. **Research Design**

   A quantitative, cross-sectional correlational research design was adopted. Data were collected at a single point in time from working professionals to determine whether training hours predict productivity levels. The design is non-experimental, as variables were not manipulated, and relies on regression analysis to estimate the direction and strength of the relationship. This design is suitable for modeling workplace outcomes where naturally occurring differences in training exposure can be analyzed statistically.

2. **Nature of the Study**

   The present study is quantitative, analytical, and explanatory in nature. It is quantitative because it relies on numerical data obtained through structured survey responses, analytical because it employs statistical techniques to examine patterns in training hours and productivity outcomes, and explanatory because it seeks to explain whether variations in training hours significantly predict changes in employee productivity. The study does not manipulate variables but observes naturally occurring conditions within workplace settings, making it suitable for modeling relationships using regression-based analysis.

3. **Source of Data**

   The study is based on primary data, which was collected directly from working professionals using a structured questionnaire. The questionnaire captured information related to demographic characteristics, total training hours completed within the past 12 months, and post-training productivity indicators. Since the dataset reflects firsthand responses rather than secondary records or published statistics, it provides original empirical evidence suitable for modeling the relationship between training hours and productivity.

## 4. Data Collection Method

Data for the study was collected using a structured online questionnaire, administered to working professionals across different roles and sectors. The questionnaire included closed-ended items related to demographic details, total training hours completed, and post-training productivity ratings. Respondents submitted their answers electronically, allowing efficient and accurate data capture. This method ensured uniformity in responses, reduced manual recording errors, and enabled the collection of quantifiable data suitable for statistical analysis.

## 5. Sampling Method and Sample Size

The study employed a non-probability convenience sampling method to select respondents, as participants were approached based on their accessibility and willingness to participate in the survey. This method was appropriate due to practical constraints and the need to collect data from diverse working professionals across different sectors within a limited time frame.

A total of 80 valid responses were obtained after data cleaning and removal of incomplete entries. These responses constitute the final sample size ($n = 80$) used for regression modeling and subsequent analysis. The sample size is adequate for basic linear regression analysis and supports the empirical examination of the relationship between training hours and productivity.

## 6. Scope of the Study

The scope of this study is limited to analyzing how employee training hours influence productivity outcomes at the individual level. The study focuses on working professionals across varied job roles and sectors, without restricting the sample to any specific industry or geographic region. The analysis is based solely on primary data collected through a structured questionnaire, and productivity is assessed using self-reported post-training ratings rather than organizational financial output. The study measures training exposure in terms of total hours completed within the past 12 months and evaluates its impact using a simple linear regression model. The scope does not extend to comparative

industry analysis, longitudinal performance tracking, or evaluation of training content, quality, or delivery modes. Instead, it concentrates on establishing a measurable statistical relationship between training hours and productivity within a cross-sectional dataset.

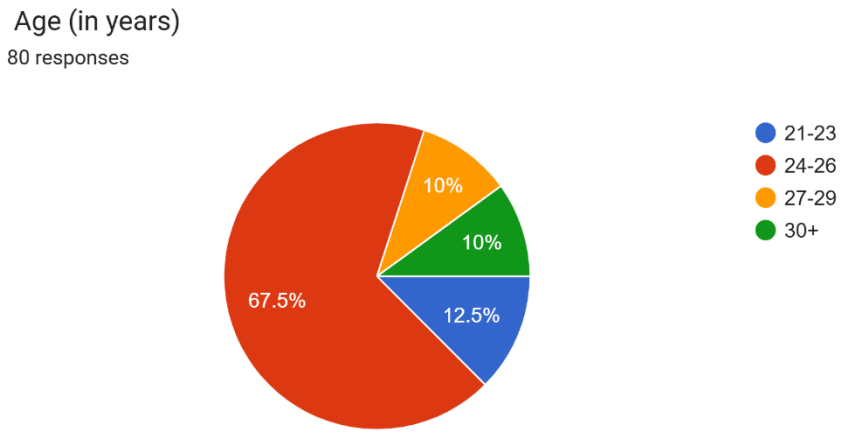
## 7. Tools and Techniques of Data Analysis

The collected data was processed and analyzed using quantitative statistical techniques. Descriptive statistics such as mean, standard deviation, and frequency distributions were utilized to summarize patterns in training hours and productivity ratings. The primary inferential analytical method employed was Simple Linear Regression to assess the predictive relationship between total training hours (independent variable) and post-training productivity ratings (dependent variable). The regression analysis was conducted under the Classical Linear Regression Model (CLRM) assumptions to ensure validity of estimation. Statistical outputs including regression coefficients, p-values, and coefficient of determination ($R^2$) were used to evaluate the strength, direction, and significance of the relationship. The analysis was performed using standard statistical software tools capable of handling regression modeling and residual diagnostics. This approach enabled empirical quantification of how variations in training exposure influence productivity outcomes.

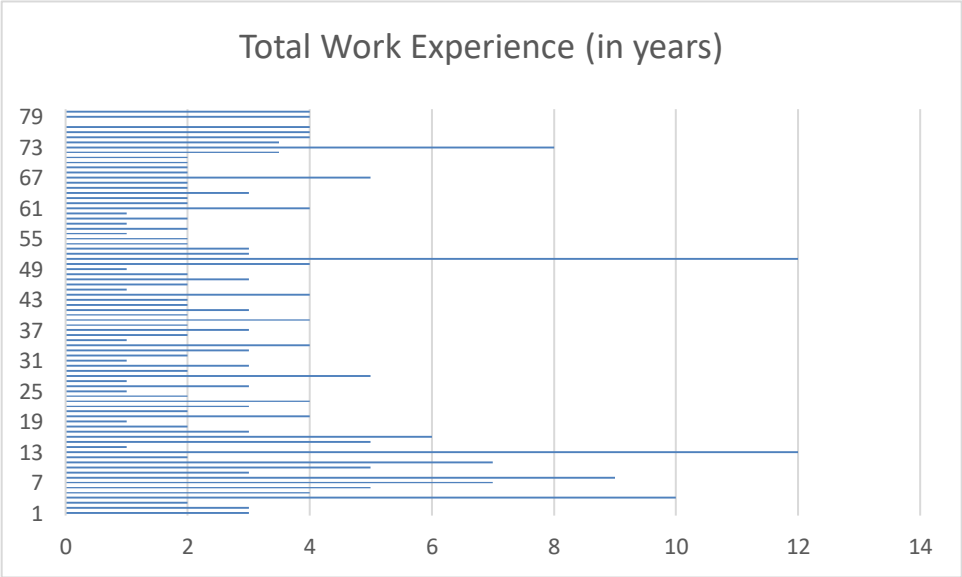
## 8. Limitations of the Study

This study is limited by its reliance on self-reported productivity ratings, which may introduce subjective bias. The sample size is relatively small and selected through convenience sampling, which may affect generalizability. The cross-sectional design captures data at a single point in time, restricting the ability to observe long-term training effects. Additionally, the study considers training hours as the sole predictor of productivity and does not account for other influencing factors such as work environment, motivation, or experience.

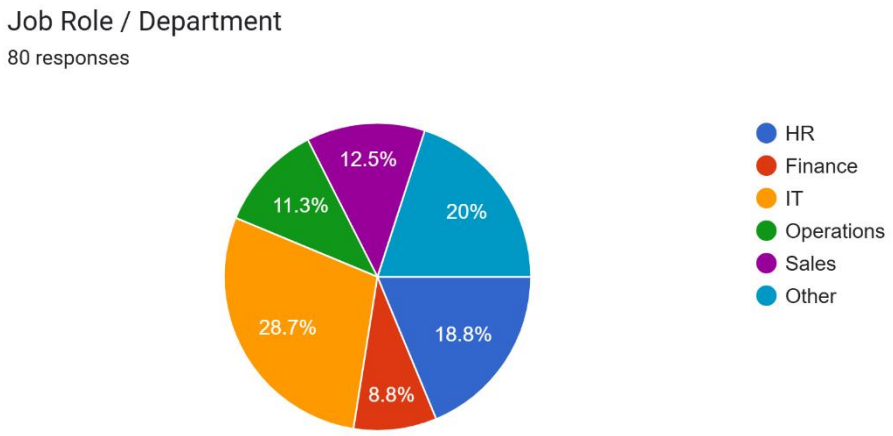
# DATA ANALYSIS



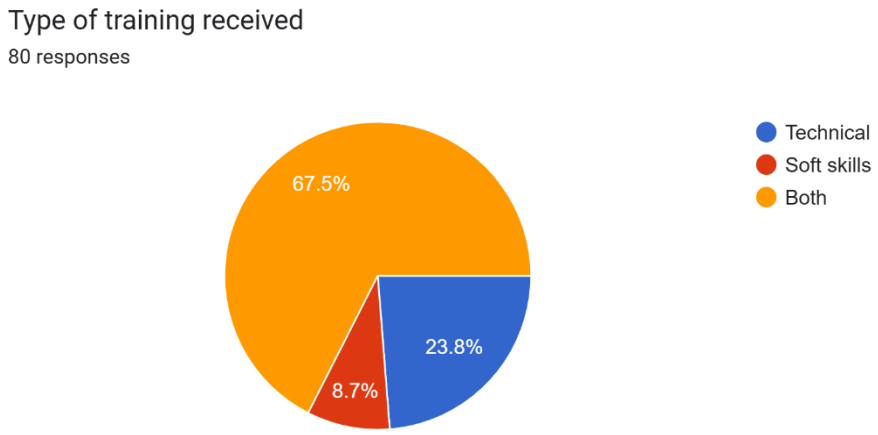

- The age distribution of respondents indicates that the majority fall within the 24–26 years category, accounting for 67.5% of the sample. This is followed by respondents aged 21–23 years (12.5%), while the 27–29 years and 30+ years categories each represent 10% of the total respondents. The distribution suggests that the dataset is predominantly composed of young working professionals in their mid-twenties, reflecting an early-career demographic typically engaged in active skill development and training participation.



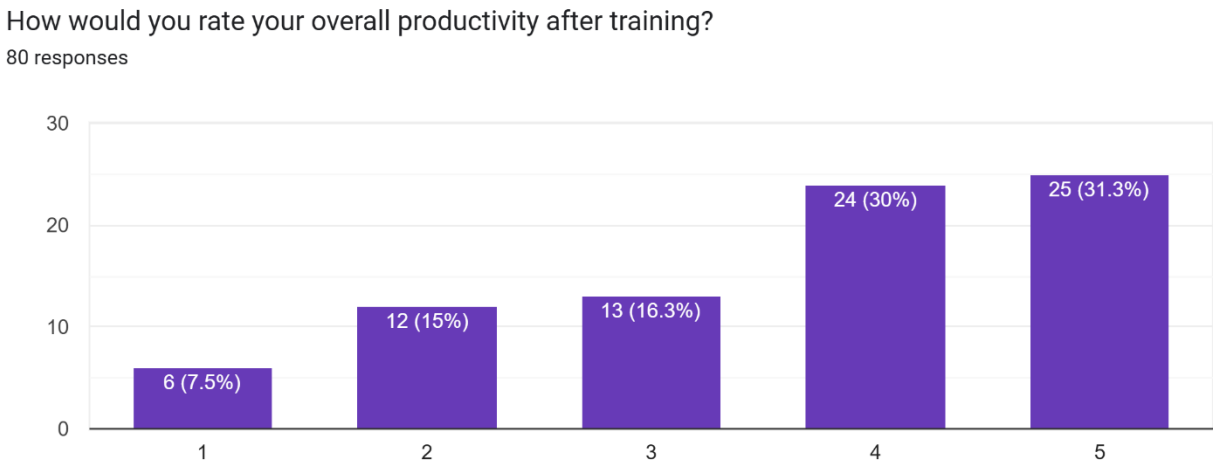
- The distribution of total work experience among respondents shows a wide spread ranging from less than 1 year to over 10 years. However, the dataset is heavily concentrated within the early-career segment, with most respondents reporting 1–4 years of experience. Only a small proportion indicate longer tenure beyond 5 years, and very few reach 10+ years. This indicates that the sample predominantly represents employees in the early stages of their professional careers, which aligns with the age distribution and suggests a workforce likely to be actively engaged in training and skill development activities.



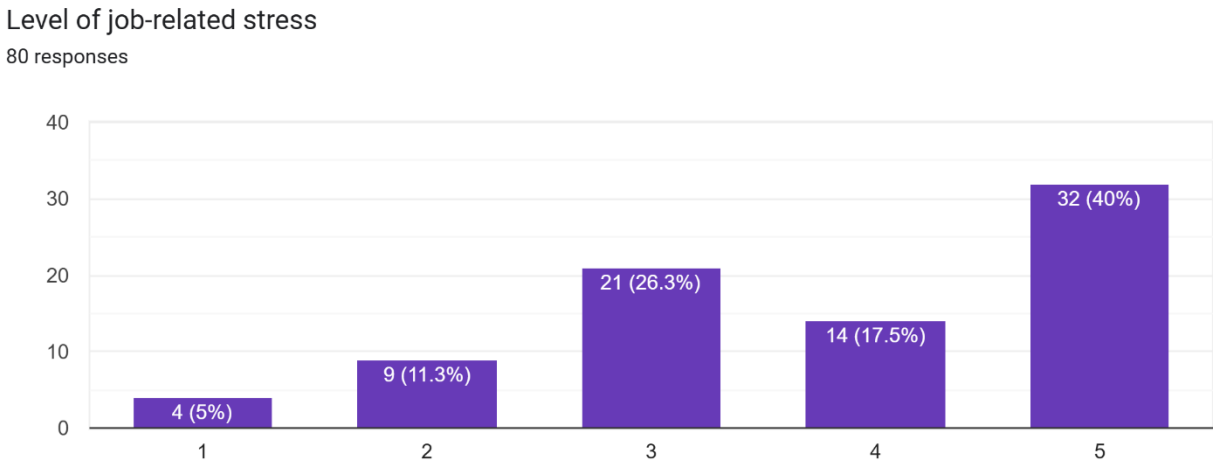

- The job role distribution reveals participation from multiple functional areas, reflecting a diverse sample of working professionals. The largest proportion of respondents belong to the IT department (28.7%), followed by HR (20%), and Other roles (18.8%). Smaller proportions are observed in Sales (12.5%), Operations (11.3%), and Finance (8.8%). This indicates that while the dataset is predominantly represented by IT and HR roles—both of which often emphasize structured training and skill development—it maintains cross-functional representation, supporting broader generalization of findings across job categories.




- The distribution of training types indicates that a majority of respondents (67.5%) received both technical and soft skills training, reflecting a balanced approach to skill development. A smaller proportion (23.8%) reported undergoing exclusively technical training, while only 8.7% received solely soft skills training. This pattern suggests that organizations increasingly adopt integrated training strategies that combine technical competencies with interpersonal and behavioral skills, aligning employee development with modern workplace requirements.



- The distribution of post-training productivity ratings shows that the majority of respondents perceived a positive improvement in their productivity after undergoing training. Specifically, 31.3% rated their productivity at the highest level (5), followed by 30% who rated it as 4. Moderate ratings (3) accounted for 16.3%, while lower ratings of 2 and 1 represented 15% and 7.5%, respectively. This pattern indicates that over 61% of respondents reported high productivity levels (ratings 4 or 5), suggesting that training interventions are generally viewed as effective in enhancing performance. The relatively smaller share of low ratings further reinforces the perceived beneficial impact of training on productivity outcomes among employees.




- The distribution of job-related stress levels indicates that a significant portion of respondents experience elevated stress at work. The highest proportion (40%) rated their stress at the maximum level (5), followed by 26.3% at level 3, and 17.5% at level 4. Lower stress ratings were comparatively less frequent, with only 11.3% selecting level 2 and 5% selecting level 1. Overall, nearly 84% of respondents reported moderate to high stress levels (ratings 3, 4, or 5), suggesting that job demands, workload, or workplace conditions may contribute to substantial stress among employees. This context is useful for understanding productivity outcomes, as high stress can influence performance, motivation, and the perceived value of training.

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## STATISTICAL ANALYSIS

A simple linear regression analysis was conducted to examine the predictive relationship between total employee training hours (independent variable) and post-training productivity ratings (dependent variable) under the Classical Linear Regression Model (CLRM) framework. The model summary shows an $R^2$ value of 0.0161, indicating that only 1.61% of the variance in productivity ratings can be explained by differences in training hours. The ANOVA results yielded an F-statistic of 1.2730 with a significance value of $p = 0.2627$, which exceeds the 0.05 threshold, indicating that the overall regression model is not statistically significant at the 5% level. The estimated regression equation was found to be $Y=3.585955+0.000286X$, suggesting a very small positive slope coefficient for training hours.

SUMMARY OUTPUT

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.126724 |
| R Square | 0.016059 |
| Adjusted R | 0.003444 |
| Standard E | 1.274415 |
| Observatio | 80 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *ignificance F* |
|---|---|---|---|---|---|
| Regressio | 1 | 2.067591 | 2.067591 | 1.273043 | 0.262655 |
| Residual | 78 | 126.6824 | 1.624133 | | |
| Total | 79 | 128.75 | | | |

| | *Coefficients* | *andard Err* | *t Stat* | *P-value* | *Lower 95%* | *Upper 95%* | *ower 95.0%* | *pper 95.0%* |
|---|---|---|---|---|---|---|---|---|
| Intercept | 3.585955 | 0.146626 | 24.45648 | 2.52E-38 | 3.294045 | 3.877865 | 3.294045 | 3.877865 |
| X Variable | 0.000286 | 0.000254 | 1.128292 | 0.262655 | -0.00022 | 0.000791 | -0.00022 | 0.000791 |

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## INTERPRETATION

The regression results indicate that while training hours have a positive coefficient, the relationship between training duration and productivity improvement is weak and statistically insignificant within the sample. With a p-value above 0.05 for both the slope and the overall model, the findings suggest that training hours alone do not meaningfully predict productivity outcomes among respondents. This implies that productivity improvements may depend more on other variables such as training quality, job relevance, motivation, workplace conditions, or organizational support, rather than the sheer number of hours spent in training. Therefore, within the context of this dataset, increased training exposure does not guarantee higher productivity, highlighting the importance of focusing on training effectiveness rather than quantity.

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## DISCUSSION

The findings of this study provide useful insights into the nuanced relationship between employee training hours and productivity outcomes. While training is widely recognized as a mechanism for enhancing employee performance, the regression analysis conducted in this study reveals that training hours alone do not serve as a strong predictor of productivity. The weak explanatory power of the regression model ($R^2 = 0.0161$) suggests that variations in productivity are influenced by additional factors beyond mere training duration. This aligns with contemporary HR and organizational behavior literature which argues that training effectiveness is contingent upon multiple dimensions, including the relevance and quality of training content, alignment with job roles, learning motivation, managerial support, and opportunities for application in the workplace.

The positive but statistically insignificant coefficient for training hours indicates that, although there may be a directional benefit associated with training exposure, the marginal impact of incremental training hours is minimal without complementary organizational conditions. The demographic characteristics of the sample—primarily early-career professionals—may further influence this outcome, as productivity for this group can be strongly shaped by factors such as job learning curves, task complexity, stress levels, workplace culture, and role clarity. The data collected on stress levels, for instance, show that a considerable proportion of respondents experience high job-related stress, which can dampen productivity irrespective of training interventions.

Additionally, the distribution of training types within the sample reveals that most respondents received a blend of technical and soft skills training. However, mixed training alone does not ensure performance improvement unless the learned skills are translated into job-relevant capabilities. This highlights a key point emphasized in prior research: training must be not only delivered, but also absorbed, contextualized, and reinforced to yield measurable productivity gains. It is therefore plausible that organizations represented in this dataset may emphasize training quantity or compliance-oriented learning rather than strategic, performance-driven learning pathways.

The discussion also acknowledges methodological considerations. Given the cross-sectional design and reliance on self-reported productivity ratings, productivity improvements may not be immediately observable, especially for skills that require sustained practice or post-training integration. Moreover, productivity as a construct is multidimensional; capturing only its subjective self-rating may omit operational performance metrics such as output quality, task efficiency, or goal attainment percentages. Hence, the insignificant statistical association observed in this study should not be interpreted as evidence against training, but rather as evidence that training must be holistically implemented and contextually measured to demonstrate tangible productivity gains.

Overall, the discussion underscores that employee development is multifaceted, and training hours represent just one input within a broader performance ecosystem. The study's results align with modern organizational perspectives emphasizing that training must be purposeful, job-aligned, and supported by conducive work environments to drive productivity. Future research incorporating variables such as training quality, relevance, motivation, work environment, managerial support, and longitudinal productivity measures may offer stronger explanatory insights.

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## CONCLUSION

This study set out to examine whether employee training hours serve as a significant predictor of post-training productivity using a Classical Linear Regression Model (CLRM) framework. While training is widely viewed as an essential component of human capital development, the statistical results from this research indicate that training hours alone have a weak and statistically insignificant effect on productivity outcomes within the sampled population. The regression model yielded a very low explanatory power ($R^2 = 0.0161$), and the slope coefficient for training hours was not statistically significant at the 5% level, suggesting that variations in productivity are driven by additional organizational, personal, and contextual factors beyond mere training duration. These findings highlight that the quality, relevance, and application of training may be more critical determinants of performance than the volume of hours spent in training. Although the analysis does not diminish the broader value of training as an organizational investment, it emphasizes the importance of adopting a more holistic approach to employee development—one that integrates structured training with supportive work environments, role clarity, motivation, and opportunities for skill application. In conclusion, the study contributes empirical evidence to the ongoing discourse on training effectiveness and underscores the need for future research to incorporate additional predictors and longitudinal designs to better capture the multifaceted impact of training on productivity.

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# LIMITATIONS & SCOPE FOR FUTURE RESEARCH

## Limitations:

This study has certain limitations that should be acknowledged when interpreting the findings. First, the analysis relies on self-reported productivity ratings, which may introduce subjective bias and may not fully capture objective performance outcomes. Second, the study utilizes a cross-sectional design, collecting data at a single point in time, which restricts the ability to observe longitudinal impacts of training on productivity. Third, the sample size is relatively small (n = 80) and drawn through convenience sampling, limiting generalizability across broader populations or industries. Additionally, training was measured solely in terms of total hours completed, without considering the qualitative dimensions of training such as content relevance, instructional quality, or alignment with job requirements. Other influential factors—such as motivation, workplace culture, managerial support, and stress levels—were not incorporated into the regression model, which may contribute to unexplained variance in productivity outcomes.

## Scope for Future Research:

Future research may expand upon the present study in several meaningful directions. Incorporating objective productivity metrics—such as task completion rates, output measures, efficiency scores, or performance evaluations—can provide a more comprehensive understanding of training outcomes. Longitudinal or panel-based research designs can help examine how training effects evolve over time, especially for skills that require sustained practice. Future studies may also explore the qualitative dimensions of training, including delivery modes, content relevance, learner engagement, and post-training reinforcement mechanisms, to better assess training effectiveness. Expanding the variable set to include motivational factors, organizational climate, job stress, or managerial support could improve model explanatory power and reveal interaction effects between training and contextual variables. Finally, sector-specific or cross-industry comparative studies with larger and probabilistic samples may help validate findings across different work environments and increase the generalizability of results.

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